SEAT No.__________

[33]

SARDAR PATEL UNIVERSITY

BCA Semester-3 On-Demand Examination [NC], February- 2024.

US03SBCA51 || DIGITAL COMPUTER ELECTRONICS

Date: 20/02/2024

Time : 02.00 p.m. – 04.00 p.m. **Marks: 50**

Q-1 Select the appropriate option for the following questions: **10**

1) The OR gate has two or more input signals. If any input is ________, the output is high.
   A. High C. Both A and B
   B. Low D. None

2) De Morgan's second theorem says that a NAND gate is equivalent to a ____________.
   A. bubbled AND C. OR bubbled
   B. bubbled NAND D. bubbled OR

3) In k-map, quad eliminates ________ variable.
   A. One C. Two
   B. Three D. Four

4) A ______ is a combinational circuit that converts binary information from the 2^n coded inputs to n outputs.
   A. Half Adder C. Encoder
   B. Decoder D. Comparator

5) A combinational circuit that performs the arithmetic addition of two bits is called_____.
   A. full adder C. binary adder
   B. half adder D. decoder

6) A multiplexer also called a __________.
   A. data multiplier C. data inverter
   B. data selector D. data remover

7) In shift right register, the arrival of the first rising clock edge sets the ______ flip-flop.
   A. Left C. Up
   B. Right D. Down

8) A ________ register is the simplest kind of register; all it does store a digital word.
   A. Shift left C. Buffer
   B. Shift Right D. Simple

9) In D flip-flop, when CLK is low then input is _______.
   A. High C. Don't care
   B. Low D. No change

10) The relationship between a function and its binary variables can be represented in ______.
   A. Truthtable C. Encoder
   B. Decoder D. Multiplexer

Q-2 Give short answers of the following questions: (ANY Eight) **16**

1) Explain NOT, XOR gate.
2) Write truth table for : A'B+B'C
3) Define comparator in short.
4) Define Flip flop. What is Race condition?

(P.T.O.)

5) What is De-Multiplexer?

6) Define parity bit. What is odd and even parity?

7) Give example of 4 bit binary subtraction.

8) What is register? Draw circuit diagram of shift right register.

9) Simplify using k-map F(A,B,C)=E(1,2,5).

10) Simplify Boolean expression and draw circuit. AB+CD+AB+CD.

Q3	Explain Basic gates.	**06**
	OR	
Q3	Explain de morgan's first and second theorem.	**06**
Q4	What is k-map? Explain pair and quad with example.	**06**
	OR	
Q4	Explain 8x3 line encoder in detail.	**06**
Q5	Explain 4*1 Multiplexer.	**06**
	OR	
Q5	Explain Half adder and Full adder in detail.	**06**
Q6	Explain controlled buffer register.	**06**
	OR	
Q6	Explain Ring Counter.	**06**

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